Reg. No. 71980900004

SRI RAMAKRISHNA INSTITUTE OF TECHNOLOGY

[Educational Service : SNR Sons Charitable Trust]
[Autonomous Institution, Reaccredited by NAAC with 'A' Grade]
[Approved by AICTE, New Delhi, Permanently Affiliated to Anna University, Chennai]
Coimbatore - 641 010.

Department of Computer Science and Engineering

Internal Test – I

Date	03.08.2026	Department	CSE		
Semester	VII	Class/section	A B & C		
Duration	100 minutes	Maximum marks	50	Regulation	R2020
Course Code & Title		20CSP36 Internet of Things			

Course Outcomes Addressed:
CO1: Ability to understand the basic concepts related to Internet of things.
CO2: Ability to apply the IoT design methodologies in IoT systems.

S. No	Questions	Cognitive Level/CO
	PART – A (Answer all questions) (5×2 =10 Marks)	
1	A farmer wants to monitor soil moisture and control irrigation using a smartphone from anywhere. Identify the technology used in this application and define it.	C1/CO1
2	List five characteristics of IoT.	C1/CO1
3	What is MQTT and write its uses.	C1/CO1
4	A logistics company continuously collects sensor data from delivery trucks and forwards it to a cloud analytics server without expecting an immediate response. Identify the communication model used and draw its architecture.	C2/CO2
5	Name the three types of services provided by cloud computing technologies.	C1/CO1
	PART – B (Answer any three questions) (3×8 =24 Marks)	**Cognitive Level/CO**
6	A smart healthcare system consists of wearable sensors, Wi-Fi gateways, cloud servers, and mobile applications. Data is securely transmitted from patients to doctors through multiple communication protocols. Explain the IoT protocols used at different layers of the IoT architecture with a neat layered diagram suitable for this healthcare application. **(NPTEL)**	C3/CO1
7	A smart parking system allows users to: View vacant parking slots using a mobile application. Receive real-time updates whenever a parking space becomes available. Explain how: RESTful-based APIs can be used for retrieving parking information. WebSocket-based APIs can be used for real-time updates.	C3/CO2
8	A smart city project monitors traffic, pollution, weather, and public safety using	C3/CO2

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	thousands of sensors connected through cloud infrastructure. The collected data is processed using big data analytics and artificial intelligence to support decision making. Identify the suitable IoT Level (Level 5 or Level 6) and explain the architecture, data flow, and working with neat diagrams.	
9.	A food delivery company develops two communication mechanisms: Customers request delivery status through a mobile application. Delivery executives continuously send their GPS location to a monitoring server. Identify the suitable communication models and illustrate them with neat sketches: Request–Response Push–Pull	C3/CO2
	PART – C (Answer any one question) (1×16 =16 Marks)	
10.	A company is designing a wearable health monitoring device that measures heart rate, body temperature, and blood oxygen level. The collected data is transmitted to the cloud for analysis and alerts. Explain the generic block diagram of the IoT device suitable for this application with a neat sketch. **(NPTEL)**	C3/CO1
11.	Consider the following IoT applications: (a) A smart home where users control lights, fans, and security cameras using a mobile application. (b) A smart irrigation system where soil moisture sensors automatically control water pumps and upload data to cloud storage. Identify the suitable IoT level for each application and justify your answer with architecture.	C3/CO2

S.No.	Cognitive Level addressed	Marks	Percentage
1.	Remember (C1)	8	11%
2.	Understand (C2)	2	3%
3.	Apply (C3)	64	86%
4.	Analyze (C4)	-	-
5.	Evaluate (C5)	-	-
6.	Create (C6)	-	-
	Total	74	100

M. Senthamilselvi 24/7/26
Course Instructor

24/7/26
Division Head

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Program Assessment Committee

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